

A 2×2 self-adjoint counterexample to the literal symmetrized-AGM norm conjecture

Autonomous mathematical discovery packet
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Status and scope. Candidate full counterexample to the operator/self-adjoint formulation written in arXiv:1803.02435, likely valid, pending human review. Three pairwise nonproportional invertible real symmetric 2×2 matrices violate the constant-one norm comparison already for $n = k = 3$. The original Recht–Ré conjecture explicitly assumes positive semidefinite matrices; that narrower problem is *not* resolved here.

1 The source question and its scope ambiguity

For operators $A_1, \dots, A_n$ and $k \leq n$, Albar, Junge, and Zhao [1] write

$$E_{\text{wo},k}(\mathcal{A}) = \frac{(n-k)!}{n!} \sum_{j_1, \dots, j_k \text{ all distinct}} A_{j_1}^* \cdots A_{j_k}^* A_{j_k} \cdots A_{j_1},$$

$$E_{\text{wr},k}(\mathcal{A}) = \frac{1}{n^k} \sum_{j_1, \dots, j_k=1}^n A_{j_1}^* \cdots A_{j_k}^* A_{j_k} \cdots A_{j_1}.$$

They ask whether

$$\|E_{\text{wo},k}(\mathcal{A})\| \leq \|E_{\text{wr},k}(\mathcal{A})\| \tag{1}$$

always holds, and state that this norm conjecture remains open.

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Ré and Recht also asked for norm or order inequalities of the following:

$$(1.7) \quad \|\mathbb{E}_{\text{wo},k}(\mathcal{A})\| \stackrel{?}{\leq} C_3 \cdot \|\mathbb{E}_{\text{wr},k}(\mathcal{A})\|,$$

$$(1.8) \quad \mathbb{E}_{\text{wo},k}(\mathcal{A}) \stackrel{?}{\leq} C_4 \cdot \mathbb{E}_{\text{wr},k}(\mathcal{A}).$$

The norm symmetric AGM conjecture asks for (1.7) with $C_3 = 1$. This still remains open.

Figure 1: The constant-one norm conjecture in arXiv:1803.02435, page 3.

There are two natural readings. Equations (1.5)–(1.7) in [1] use adjoints and operator variables, and the subsequent results speak of families of operators. Read literally in that generality, (1) is false by the theorem below. But the preceding motivation in the same paper starts with positive definite matrices, as the following source excerpt shows.

More decisively, Recht and Ré’s original weak-variance conjecture explicitly assumes positive semidefinite matrices [2, Conjecture 1]. Our result therefore resolves the literal operator version and isolates positivity as an indispensable hypothesis; it does not claim to settle the historical PSD version.

It's natural to ask whether the MacLaurin inequalities still hold if we replace the real numbers with positive definite matrices. Namely, if $A_1, \dots, A_n$ are a collection of positive definite operators in some Hilbert space, the MacLaurin mean is defined as

$$(1.1) \quad \mathcal{P}_d(\mathcal{A}) = \frac{(n-d)!}{n!} \sum_{1 \leq j_1, \dots, j_d \leq n \text{ all different}} A_{j_1} \cdots A_{j_d}$$

It is still open whether the norm of $\mathcal{P}_d$ is always less than that of $(\mathcal{P}_1)^n$, i.e.

$$(1.2) \quad \|\mathcal{P}_d(\mathcal{A})\| \stackrel{?}{\leq} C_1 \cdot \|\mathcal{P}_1(\mathcal{A})\|^d$$

Figure 2: The positive-definite motivating context in arXiv:1803.02435, page 2.

2 The counterexample

Theorem 1. *The norm comparison (1) fails for $n = k = 3$ on the real two-dimensional Hilbert space, even when A_1, A_2, A_3 are pairwise nonproportional invertible self-adjoint operators. One may take*

$$A_1 = \begin{pmatrix} -1 & -2 \\ -2 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} -2 & 2 \\ 2 & 1 \end{pmatrix}, \quad A_3 = \begin{pmatrix} -1 & 1 \\ 1 & 2 \end{pmatrix}. \quad (2)$$

For this triple,

$$E_{\text{wo},3} = \begin{pmatrix} 521/6 & 27 \\ 27 & 641/6 \end{pmatrix}, \quad E_{\text{wr},3} = \frac{1}{27} \begin{pmatrix} 3081 & 302 \\ 302 & 3083 \end{pmatrix}, \quad (3)$$

and

$$\|E_{\text{wo},3}\| = \frac{581}{6} + \sqrt{829} > \frac{3082 + \sqrt{91205}}{27} = \|E_{\text{wr},3}\|. \quad (4)$$

The norm ratio is approximately 1.00233216996281.

Proof. All three matrices in (2) are real symmetric. Their determinants are respectively $-4, -6, -3$, so they are invertible and indefinite; inspection also shows that no two are proportional.

For a word $\mathbf{j} = (j_1, j_2, j_3)$ put $P_{\mathbf{j}} = A_{j_3} A_{j_2} A_{j_1}$. Its summand in either expectation is $P_{\mathbf{j}}^* P_{\mathbf{j}}$. Direct integer multiplication over the six permutations and over all twenty-seven words gives

$$\sum_{\mathbf{j} \text{ distinct}} P_{\mathbf{j}}^* P_{\mathbf{j}} = \begin{pmatrix} 521 & 162 \\ 162 & 641 \end{pmatrix}, \quad \sum_{\mathbf{j} \in \{1,2,3\}^3} P_{\mathbf{j}}^* P_{\mathbf{j}} = \begin{pmatrix} 3081 & 302 \\ 302 & 3083 \end{pmatrix}.$$

Division by $3! = 6$ and $3^3 = 27$ proves (3). The larger eigenvalues of these two positive matrices are exactly the quantities displayed in (4).

The strict comparison needs no decimal approximation. Since $(1439/50)^2 < 829$ and $303^2 > 91205$,

$$\|E_{\text{wo},3}\| - \|E_{\text{wr},3}\| > \frac{581}{6} + \frac{1439}{50} - \frac{3082 + 303}{27} = \frac{164}{675} > 0.$$

This proves the claimed failure. □

Remark 2 (an even shorter nonself-adjoint witness). *Let $N = e_{12}$ and take $A_1 = N$, $A_2 = 2N$, $A_3 = N^*$. Nilpotence leaves only alternating words, giving*

$$E_{\text{wo},3} = \frac{4}{3}e_{22}, \quad E_{\text{wr},3} = \frac{5}{27}e_{11} + \frac{25}{27}e_{22}.$$

Thus the arbitrary-operator norm ratio can already be $36/25$. Theorem 1 is stronger in regularity: its matrices are self-adjoint, invertible, and pairwise nonproportional.

3 What remains open

Each matrix in (2) has one positive and one negative eigenvalue. The witness therefore cannot answer the PSD problem by reinterpretation. This boundary is substantive: searches parameterizing $A_i = X_i^* X_i$ over real and complex matrices, through $n = k = 6$ and dimensions two and three, found no ratio above one; local optimization repeatedly converged to equality families. Positive block dilations and scalar shifts of the indefinite witnesses also lost the violation before reaching the positive cone. These computations are only stress tests, not evidence sufficient for an affirmative theorem.

The 2020 Lai–Lim counterexample [3] concerns the other Recht–Ré conjecture, the unsymmetrized first-moment/bias comparison for PSD matrices. It does not supply the palindromic second-moment comparison considered here. Accordingly, the exact remaining question is:

Does (1) hold when every A_i is positive semidefinite?

The present theorem shows that any affirmative result must use positivity, not merely self-adjointness, invertibility, small dimension, or the shortest nontrivial sampling length.

4 Verification and review focus

The companion script `code/exact_verifier.py` enumerates all words in exact SymPy arithmetic, checks (3), checks both eigenvalue formulas, and verifies the rational gap certificate. It is an independent reproducibility aid; the displayed proof itself is finite and exact.

A reviewer should prioritize: (i) that the word order matches equations (1.5)–(1.7) of [1]; (ii) the two integer sums in the proof; and (iii) the scope distinction between the literal operator formulation and the positive-semidefinite hypothesis in [2].

References

- [1] W. Albar, M. Junge, and M. Zhao, *On the symmetrized arithmetic-geometric mean inequality for operators*, arXiv:1803.02435 (2018).
- [2] B. Recht and C. Ré, *Beneath the valley of the noncommutative arithmetic-geometric mean inequality: conjectures, case-studies, and consequences*, COLT 2012; arXiv:1202.4184.
- [3] Z. Lai and L.-H. Lim, *Recht–Ré noncommutative arithmetic-geometric mean conjecture is false*, Proc. ICML 2020; arXiv:2006.01510.